

Spectroscopic detection of isolated carbonic acid

Tetsuya Mori, Kohsuke Suma, Yoshihiro Sumiyoshi, and Yasuki Endo

Citation: *J. Chem. Phys.* **130**, 204308 (2009); doi: 10.1063/1.3141405

View online: <http://dx.doi.org/10.1063/1.3141405>

View Table of Contents: <http://aip.scitation.org/toc/jcp/130/20>

Published by the [American Institute of Physics](#)

Spectroscopic detection of isolated carbonic acid

Tetsuya Mori, Kohsuke Suma,^{a)} Yoshihiro Sumiyoshi, and Yasuki Endo^{b)}

Department of Basic Science, Graduate School of Arts and Sciences, The University of Tokyo, Komaba, Meguro-ku, Tokyo 153-8902, Japan

(Received 14 April 2009; accepted 4 May 2009; published online 29 May 2009)

Carbonic acid (*cis-trans* H₂CO₃) in the gas phase has been successfully produced in a supersonic jet using a pulsed discharge nozzle, and pure rotational transitions of this molecule have been observed by Fourier-transform microwave spectroscopy. Although the observed *cis-trans* conformer is not the global minimum structure, it is an important conformer as a starting point of its dissociation to CO₂ and H₂O. Three deuterated isotopologues of the *cis-trans* conformer (*cis-trans* HD₂CO₃, *cis-trans* DHCO₃, and *cis-trans* D₂CO₃) have also been observed, yielding the *r*₀ structure of *cis-trans* H₂CO₃. The present result is accurate enough to be used in radio astronomical observations. © 2009 American Institute of Physics. [DOI: 10.1063/1.3141405]

I. INTRODUCTION

Carbonic acid (H₂CO₃) is a diprotic acid formed by the hydration of carbon dioxide and is playing important roles in various areas of science. For example, it works as an important buffer to stabilize the pH of the blood and other biological fluids in the form of the bicarbonate ion.¹ Carbonate minerals, which are salts of carbonic acid, are important in geology.² It is also receiving a particular attention as a prospective interstellar molecule, which could be produced in icy grain mantles.^{3–5}

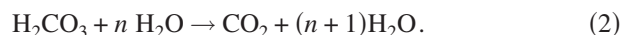
In spite of its importance in various fields of science, carbonic acid had not been detected by any spectroscopic means for a long period, although microwave spectra of the H₂O–CO₂ complex have been observed in the early 1980s.⁶ In the late 1980s, a peak at a mass-to-charge ratio, *m/z*=62 (attributed H₂CO₃⁺), had been detected by neutralization-reionization mass spectroscopic techniques during the thermal decomposition of NH₄HCO₃,⁷ which may be the first observation of stable and isolated carbonic acid. In the condensed phase, peaks, which could be attributed to the vibrational transitions of carbonic acid, were observed in infrared spectroscopic studies of H₂O+CO₂ ices exposed to high-energy radiation.^{3,4} Hage and co-workers^{5,8} reported that crystalline carbonic acid sublimates without decomposition and recondenses again under a water-free condition, confirming its stability in the gas phase.

A number of quantum chemical studies have been reported for the formation of carbonic acid from water and carbon dioxide, where the reaction



is either exothermic or endothermic depending on the levels of computations. However, recent better quality calculations have reported that this reaction is endothermic.^{9–16} A recent quantum chemical study has predicted that isolated carbonic

acid is kinetically stable; the calculated half-life of the pure carbonic acid molecule is estimated to be 0.18 million years at the room temperature [Eq. (2), with *n*=0], whereas the half-life decrease to 10 h or 2 min when catalyzed by one or two water molecules, respectively [Eq. (2), with *n*=1 or 2].¹⁴



This is due the fact that the dissociation barrier becomes lower, if more water molecules are attached to carbonic acid. Although no experimental data exist to determine its molecular structure, *ab initio* calculations have shown that carbonic acid has three conformers, *cis-cis*, *cis-trans*, and *trans-trans*^{9–12,16} as shown in Fig. 1(a). Energetically, the *cis-cis* conformer is the most stable, and the *cis-trans* conformer is the next stable conformer. The isomerization path from the *cis-cis* conformer to the *cis-trans* conformer via the change in one dihedral angle O=C–O–H has the lowest barrier. Another path via the intramolecular proton transfer from one oxygen atom to another while maintaining a planar structure has a much higher barrier.¹² It should be noted that, although the *cis-trans* conformer is not the global minimum structure, it is considered to be an important isomer as an intermediate when the carbonic acid molecule decomposes to CO₂ and H₂O.^{13,16} In other words, the *cis-cis* conformer can hardly decompose directly, requiring a conformation change to the *cis-trans* conformer through the dihedral angle change for the dissociation to CO₂+H₂O. Similarly, the water complex of carbonic acid, for which the global minimum structure is *cis-cis* H₂CO₃–H₂O, dissociates via *cis-trans* H₂CO₃–H₂O complex.^{10,13,14} Thus, *cis-trans* H₂CO₃ is an important isomer of carbonic acid for the investigations of dissociation processes, or discussions of the stability of carbonic acid.

So far, since isolated carbonic acid had not been observed by spectroscopic means to determine its structure, various theoretical assumptions had not been confirmed experimentally. In the present study, observations of pure rotational transitions of isolated *cis-trans* H₂CO₃ and its deuterated isotopologues in the gas phase are reported together with results of *ab initio* calculations at the coupled-cluster level, which is higher than those of previous studies. The

^{a)}Present address: Department of Chemical System Engineering, The University of Tokyo, Hongo 7-3-1, Bunkyo-ku, Tokyo, 113-8656, Japan.

^{b)}Author to whom correspondence should be addressed. FAX: +81-3-5454-6721. Electronic mail: endo@bunshi.c.u-tokyo.ac.jp.

TABLE I. Calculated energies (in kcal/mol) of geometry-optimized structures of respective species with respect to the *cis-cis* conformer. The optimized structures are shown in Fig. 2.

	<i>cis-cis</i>		<i>cis-trans</i>		$\text{CO}_2\text{-H}_2\text{O}$	$\text{CO}_2+\text{H}_2\text{O}$	
	H_2CO_3	TS1 ^a	H_2CO_3	TS2 ^b			
CCSD(T)/cc-pVQZ	0.00	10.47	1.71	51.29	-5.59	-2.47	Present
CCSD(T)-F12/aug-cc-pVDZ	0.00	10.53	1.74	51.28	-5.44	-2.47	Present
DFT-BLYP/PW	0.00	9.6	0.97	37.1	-9.58	...	Ref. 16
QCISD(T)/6-311++G** ^c	0.00	10.33	1.86	45.77	-9.68	...	Ref. 12

^aThe transition state structure of the isomerization between *cis-cis* and *cis-trans* conformers by a change in one dihedral angle $\text{O}=\text{C}-\text{O}-\text{H}$.

^bThe transition state structure of the dissociation from *cis-trans* conformer to the $\text{CO}_2\text{-H}_2\text{O}$ complex.

^cGeometry was optimized at MP2/6-311++G** level.

molecular structure of this isomer is determined experimentally, confirming the validity of the theoretical studies by comparing the *ab initio* structure with the experimental one.

II. AB INITIO CALCULATION

So far, a large number of theoretical investigations have been reported.⁹⁻¹⁶ We also have performed *ab initio* calculations at the coupled-cluster level of theory with single, double, and noniterative triple excitations [CCSD(T)] using the polarized valence quadruple-zeta correlation-consistent, cc-pVQZ, basis set. The optimized geometry of our calculations agrees well with previous quantum chemical studies. The optimized structures of the three conformers and the potential energy curve of the isomerization are shown in Fig. 1. Note that the left hand side and the right hand side of this curve are plotted as functions of dihedral angles $\text{O}=\text{C}-\text{O}_2-\text{H}$ (ϕ_2) and $\text{O}=\text{C}-\text{O}_1-\text{H}$ (ϕ_1), respectively, where O1 and O2 are defined as shown in Fig. 1(b), all other geometrical parameters are optimized at each dihedral angle while keeping the planarity for the heavy atoms and the other hydrogen atom. As shown in Fig. 1, the *cis-cis* conformer is the most stable, and the energies of the *cis-trans* and *trans-trans*

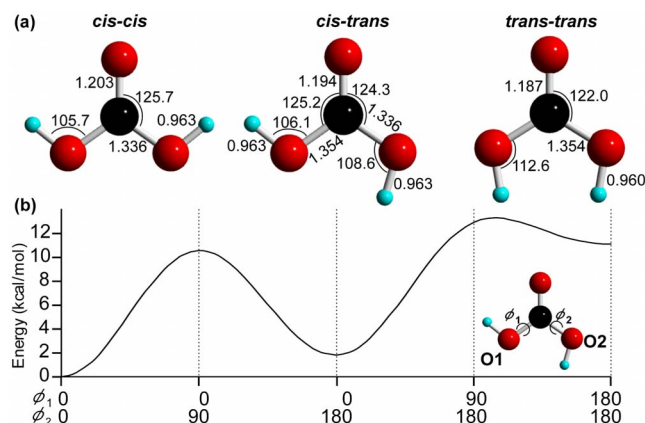


FIG. 1. (Color online) (a) The molecular structures of the three conformers of the carbonic acid molecule obtained by *ab initio* calculations at the CCSD(T)/cc-pVQZ level. Structural parameters are shown in the figure in Å and degrees. (b) Potential energy curve calculated as a function of dihedral angles, ϕ_1 and ϕ_2 . The left hand side and the right hand side show isomerizations with the changes of ϕ_2 and ϕ_1 , respectively, where $\phi_1=\phi_2=0^\circ$ corresponds to the *cis-cis* conformer, $\phi_1=180^\circ$ and $\phi_2=0^\circ$ the *cis-trans* conformer, and $\phi_1=\phi_2=180^\circ$ the *trans-trans* conformer. At each point of this curve, all other degrees of freedom are optimized in energy at the CCSD(T)/cc-pVTZ level.

trans conformers are 1.74 and 10.9 kcal/mol higher, respectively. The smallest isomerization barrier between the two lowest energy minima *cis-cis* and *cis-trans* is 10.5 kcal/mol. This barrier ensures that the *cis-trans* conformer could exist stably. Since the lowest energy *cis-cis* conformer has 2.47 kcal/mol higher energy than that of $\text{CO}_2+\text{H}_2\text{O}$, the formation reaction of carbonic acid is calculated to be endothermic in agreement with recent quantum chemical calculations.¹²⁻¹⁶

The energies of all the local minimum structures and the transition state structure related to the dissociation process of carbonic acid are calculated at the CCSD(T)/cc-pVQZ level and the explicitly correlated CCSD(T) theory (CCSD(T)-F12) with the aug-cc-pVDZ basis set augmented with diffuse functions. The calculated energies and those obtained by two recent studies are summarized in Table I, and a plot of the calculated energies with the optimized geometries are shown in Fig. 2. Our results roughly agree with those of previous studies. However, the transition state structure of dissociation, TS2 in Table I and Fig. 3, has a slightly higher energy than those in Refs. 12 and 16, where the dissociation barrier from the *cis-trans* conformer is calculated to be 49.6 kcal/mol. This result shows the isolated *cis-trans* conformer could exist stably.

The spectral search was performed based on the estimated rotational constants, which were derived by the structure optimized at the CCSD(T)/cc-pVQZ level of calculations. We examined the contribution with respect to the basis set size for the energies and the optimized geometries from cc-pVDZ to cc-pV5Z for the CCSD(T) calculations of *cis-*

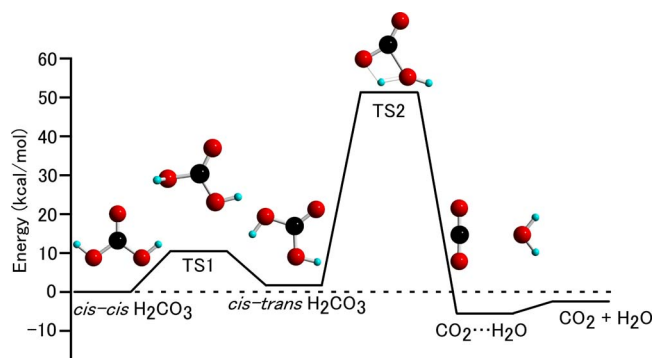


FIG. 2. (Color online) The energy levels and structures of the local minima and the transition state for the dissociation of *cis-cis* H_2CO_3 complex to $\text{CO}_2+\text{H}_2\text{O}$ [Eq. (2), with $n=0$] calculated at the CCSD(T)-F12/aug-cc-pVDZ level.

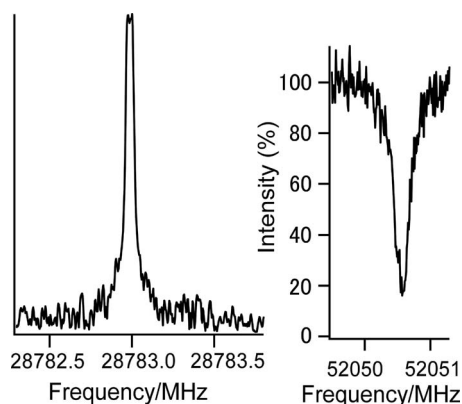


FIG. 3. (Left) The rotational transition ($J_{KaKc}=2_{02}-1_{11}$) observed by FTMW spectroscopy. This spectrum was obtained by averaging 500 free induction decay signals. (Right) The double resonance spectrum ($J_{KaKc}=2_{02}-1_{11}$), observed as a change in intensity of the FTMW spectrum shown in the left panel.

trans H_2CO_3 , showing that the cc-pVQZ basis set is sufficiently large as far as relative energies and geometries are concerned. We also performed the CCSD(T)-F12/aug-cc-pVDZ level of calculation, which gave similar results to those calculated at the CCSD(T)/cc-pVQZ level as shown in Table I. At the CCSD(T)-F12 theory, the convergence of the electron correlation energy with respect to the basis set size is dramatically improved, showing near basis set limit energy with a relatively small basis set, requiring much less computation time.¹⁷ The eQq coupling constants of the deuterons were estimated at the QCISD/cc-pVTZ level of calculation. All the calculations in the present study were performed us-

ing the MOLPRO 2006.1 and 2008.1 packages,¹⁸ except for the calculations of the hyperfine coupling constants by the QCISD method, which were performed using the GAUSSIAN 03 package.¹⁹

III. EXPERIMENT

The carbonic acid molecule was produced by an electronic discharge, induced by applying a pulsed high voltage of 1.8 kV, of a sample gas mixture within a pulsed discharge nozzle (PDN).²⁰ The sample gas mixture, 5% of CO_2 diluted with Ar, was introduced to the PDN after passing through a reservoir filled with pure water. The isotopologues whose hydrogen atoms are fully or partly deuterated are produced using the reservoir containing heavy water (D_2O) or $\text{H}_2\text{O} + \text{D}_2\text{O}$ (1:1) instead of pure water. Discharge products are then expanded into a Fabry-Pérot cavity of a Fourier-transform microwave (FTMW) spectrometer as a supersonic jet with a stagnation pressure at 3 atm, by which pure rotational transitions of carbonic acid were observed.²¹ The pressure in the vacuum chamber is kept at 4×10^{-5} Torr. One example of the observed spectra is shown in Fig. 3 (left). To extend the measurements to other possible rotational transitions and thereby confirm carbonic acid as the spectral carrier, we used a double resonance technique.²² In this technique, millimeter-wave radiation was sent into the Fabry-Pérot cavity of the FTMW spectrometer while monitoring a centimeter-wave transition by the FTMW system, and transitions induced by the millimeter-wave radiation were observed as a change in intensity of the monitor transition. An

TABLE II. Observed transition frequencies of *cis-trans* H_2CO_3 (in MHz).

J'	K'_a	K'_c	J''	K''_a	K''_c	Obs.	Obs.-Calc.
1	0	1	0	0	0	17 215.1941	0.0085
1	1	1	0	0	0	17 570.7500	0.0009
1	1	0	1	0	1	5 986.5656	-0.0122
1	1	0	1	1	1	5 630.9969	-0.0175
2	1	1	1	1	0	40 061.1864	-0.0034
2	2	1	1	1	0	41 127.8513	-0.0080
2	0	2	1	0	1	29 138.5484	-0.0018
2	1	2	1	0	1	29 154.8563	0.0061
2	2	0	1	1	1	52 050.5693	0.0121
2	0	2	1	1	1	28 782.9773	-0.0094
2	1	2	1	1	1	28 799.2836	-0.0032
2	2	0	2	1	1	6 358.3523	-0.0007
2	2	0	2	2	1	5 291.6711	-0.0124
2	1	1	2	0	2	16 909.2223	0.0048
2	2	1	2	1	2	17 959.5836	-0.0033
3	2	1	2	2	0	62 723.0224	-0.0012
3	3	1	2	2	0	64 855.8303	-0.0026
3	0	3	2	0	2	40 567.0024	-0.0146
3	1	3	2	0	2	40 567.4748	-0.0092
3	0	3	2	1	2	40 550.7298	0.0128
3	1	3	2	1	2	40 551.2006	0.0167
3	3	0	3	2	1	6 946.1031	0.0031
3	3	0	3	3	1	4 813.2930	0.0023
3	1	2	3	0	3	28 972.6328	-0.0012
3	2	2	3	1	3	29 053.4422	0.0050

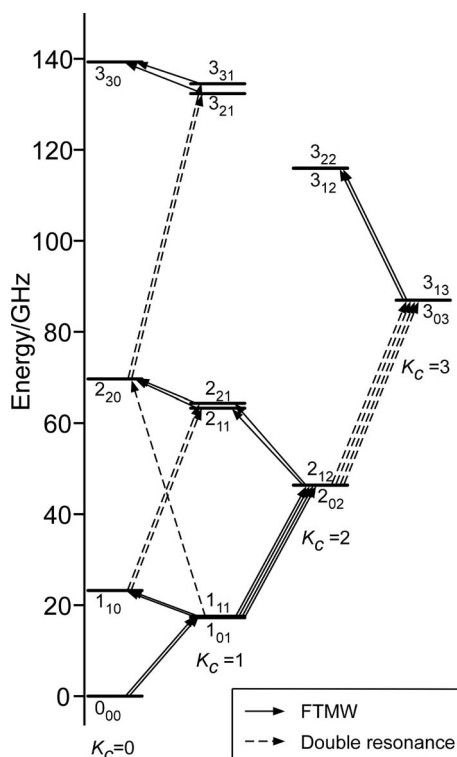


FIG. 4. The rotational energy levels and the observed transitions of *cis-trans* H₂CO₃. Solid arrows and dashed arrows indicate transitions observed by the FTMW spectrometer and the double resonance technique, respectively (see text for details).

example of the double resonance signal is also shown in Fig. 3 (right), where the transition shown in the left was used as the monitor.

IV. RESULTS AND DISCUSSION

A total of 25 transitions, both *a*-type ($\Delta K_a=0$ and $\Delta K_c=\pm 1$) and *b*-type ($\Delta K_a=\pm 1$ and $\Delta K_c=\pm 1$) in the frequency region from 4.8 to 65 GHz, were observed. All the observed transition frequencies are listed in Table II. The rotational energy levels of carbonic acid with transitions observed by the two spectroscopic methods are summarized in Fig. 4. All the observed transitions were subjected to a least-

squares analysis using Watson's *A*-reduced Hamiltonian including the quartic centrifugal distortion terms,²³

$$H = \frac{A+B}{2} \hat{J}^2 + \left(C - \frac{A+B}{2} \right) \hat{J}_z^2 + \frac{A-B}{4} (\hat{J}_+^2 + \hat{J}_-^2) - \Delta_J \hat{J}^4 - \Delta_{JK} \hat{J}^2 \hat{J}_z^2 - \Delta_K \hat{J}_z^4 - \frac{1}{2} \{ \delta_J \hat{J}^2 + \delta_K \hat{J}_z^2, \hat{J}_+^2 + \hat{J}_-^2 \}, \quad (3)$$

where $\{A, B\} = AB + BA$. The experimentally determined molecular constants are listed in Table VI. The standard deviation of the fit is 10.5 kHz, which is reasonable compared to the experimental accuracies of measurements. Only the *b* axis (*C*₂-symmetry axis) component of the dipole moment is nonzero for the *cis-cis* and the *trans-trans* conformers because of their *C*_{2v} symmetry, while both *a* and *b* axis components are nonzero for the *cis-trans* conformer, which are calculated to be $\mu_a=1.0$ D and $\mu_b=2.9$ D at CCSD(T)/cc-pVQZ. Since both *a*-type and *b*-type transitions have been observed, and the determined rotational constants agree well with those of *cis-trans* H₂CO₃ predicted by the *ab initio* calculations, we concluded that *cis-trans* H₂CO₃ was the spectral carrier. In this study, we scanned the 29 and 17 GHz regions, where both *cis-cis* and *cis-trans* H₂CO₃ have *J*=2–1 and *J*=1–0 transitions. However, all the observed lines were assigned to the *cis-trans* conformer. No spectrum of the *cis-cis* conformer has been observed yet, although it is the global minimum conformer, mainly due to its small dipole moment of $\mu_b=0.2$ D.

In addition to the transitions of *cis-trans* H₂CO₃, 14 rotational transitions of *cis-trans* HDCO₃ (H atom is at the *cis* position and D atom is at the *trans* position), 11 rotational transitions of *cis-trans* DHCO₃ (D atom is at the *cis* position and H atom is at the *trans* position), and 11 rotational transitions of *cis-trans* D₂CO₃ were observed using D₂O or D₂O+H₂O (1:1) instead of H₂O. The observed lines are listed in Tables III–V. It has been predicted by our *ab initio* calculations that *cis-trans* D₂CO₃ has a small *a* axis component of the dipole moment ($\mu_a=0.16$ D, while $\mu_b=3.1$ D) because of a small rotation of the inertial axis system caused by the deuterium substitution. In fact, only one *a*-type transition was observed by the double resonance technique using

TABLE III. Observed transition frequencies of *cis-trans* HDCO₃ (in MHz).

<i>J'</i>	<i>K'_a</i>	<i>K'_c</i>	...	<i>J''</i>	<i>K''_a</i>	<i>K''_c</i>	Obs.	Obs.-Calc.
1	0	1	...	0	0	0	16 263.6930	−0.0074
1	1	1	...	0	0	0	17 354.6484	−0.0068
1	1	0	...	1	0	1	6 174.3773	−0.0039
2	0	2	...	1	0	1	28 377.4227	0.0202
2	1	2	...	1	0	1	28 534.8625	−0.0044
2	0	2	...	1	1	1	27 286.4422	−0.0055
2	1	2	...	1	1	1	27 443.9133	0.0011
2	2	0	...	2	1	1	7 422.7047	−0.0022
2	1	1	...	2	0	2	15 407.6211	0.0030
2	2	1	...	2	1	2	18 522.9797	0.0009
3	0	3	...	2	0	2	39 220.4810	0.0017
3	1	3	...	2	0	2	39 234.8660	0.0046
3	0	3	...	2	1	2	39 063.0227	0.0078
3	1	3	...	2	1	2	39 077.3800	−0.0170

TABLE IV. Observed transition frequencies of *cis-trans* DHCO₃ (in MHz).

J'	K'_a	K'_c		J''	K''_a	K''_c	Obs.	Obs.-Calc.
1	0	1	...	0	0	0	16 333.9008	-0.0123
1	1	1	...	0	0	0	17 185.4063	0.0049
1	1	0	...	1	0	1	6 031.3359	-0.0049
2	0	2	...	1	0	1	28 242.8242	0.0062
2	1	2	...	1	0	1	28 339.3914	-0.0058
2	2	0	...	1	0	1	50 858.1493	0.0028
2	0	2	...	1	1	1	27 391.3305	0.0008
2	1	2	...	1	1	1	27 487.9109	0.0020
2	2	0	...	2	1	1	6 979.3141	-0.0034
3	0	3	...	2	1	2	39 003.4444	-0.0084
3	1	3	...	2	1	2	39 010.3582	0.0069

a microwave power amplifier for the pump radiation to compensate the small dipole moment. Nuclear quadrupole hyperfine structures caused by the nuclear spins of the deuterons were observed in the spectra of deuterated species. We used the frequencies of the strongest peaks for the least square analysis because the splittings are comparable to the linewidths of the spectra. In fact, estimated nuclear quadrupole coupling constant, eQq , of *cis-trans* D₂CO₃ are 0.31 and 0.03 MHz for the deuterons at the *cis* position and the *trans* position, respectively. The hyperfine splittings could be resolved by a reduction of the linewidth by the change in the nozzle arrangement to make the molecular beam parallel to the FTMW cavity,²⁴ although the perpendicular beam setup was adopted in the present study. The rotational constants of the four isotopologues determined by the least-squares fittings of the experimental data are listed in Table VI along with those predicted by the *ab initio* calculations. In these least-squares fittings, δ_J and δ_K of three deuterated isotopologues are fixed to the values of *cis-trans* H₂CO₃. Differences between the experimental and *ab initio* rotational constants are less than 0.6% for all the isotopologues.

Planarity of *cis-trans* H₂CO₃ is confirmed by small positive values of their inertial defects, i.e., $\Delta I = I_C - I_A - I_B = 0.106 \text{ u}\text{\AA}^2$ for *cis-trans* H₂CO₃, $0.099 \text{ u}\text{\AA}^2$ for *cis-trans* HDCO₃, $0.101 \text{ u}\text{\AA}^2$ for *cis-trans* DHCO₃, and $0.095 \text{ u}\text{\AA}^2$ for *cis-trans* D₂CO₃, as expected for planar molecules. The r_0 structure of *cis-trans* H₂CO₃ was determined by a least-squares fitting of the determined rotational constants A and B of the four isotopologues assuming their planarity. In this

procedure, the C–O–H/D angles and the OD bond length were fixed to the *ab initio* values shown in Fig. 1, and the OH bond length was fixed to be 0.005 Å longer than the *ab initio* value to take into account the OH zero point vibration.²⁵ The r_0 structural parameters are determined as $r(\text{C}=\text{O})=1.188 \text{ \AA}$, $r(\text{C}-\text{O}_{\text{cis}})=1.345 \text{ \AA}$, $r(\text{C}-\text{O}_{\text{trans}})=1.357 \text{ \AA}$, $\angle(\text{O}=\text{C}-\text{O}_{\text{cis}})=126.8^\circ$, and $\angle(\text{O}=\text{C}-\text{O}_{\text{trans}})=122.9^\circ$ with the standard deviation of the fit to be 0.4 MHz, where O_{cis} and O_{trans} mean O atoms which are bonded to the H atoms at the *cis* and *trans* positions, respectively. The r_0 and *ab initio* structural parameters are summarized in Table VII. The determined structure has similar structural constants to that of formic acid, $r(\text{C}=\text{O})=1.202 \text{ \AA}$, $r(\text{C}-\text{O})=1.343 \text{ \AA}$, and $\angle(\text{O}=\text{C}-\text{O})=124.9^\circ$.²⁶

In interstellar chemistry, recent observations, experiments, and models of grain-surface chemistry have shown that complex and saturated molecules are produced mainly in icy grain mantles rather than in the gas phase, and they are released into the gas phase when the ices evaporate.^{4,27–29} For example, both gas and solid formic acid (HCOOH), which is considered to be produced in an H₂O+CO ice exposed to high-energy radiation, has been observed in various astronomical sources.^{27,30–32} In addition, it is suggested that prebiotic molecules such as glycine are also produced in icy grain mantles.²⁸ Carbonic acid has been confirmed to be produced in a H₂O+CO₂ ice irradiated by high-energy radiation,^{3,4} and H₂O and CO₂ have been observed in various interstellar objects in both gas and solid phases.³³ It is now

TABLE V. Observed transition frequencies of *cis-trans* D₂CO₃ (in MHz).

J'	K'_a	K'_c		J''	K''_a	K''_c	Obs.	Obs.-Calc.
1	0	1	...	0	0	0	15 707.7203	0.0006
1	1	1	...	0	0	0	16 677.6344	0.0034
1	1	0	...	1	0	1	5 904.2320	0.0016
2	1	2	...	1	0	1	27 450.9656	-0.0064
2	0	2	...	1	1	1	26 351.6461	0.0011
2	2	0	...	1	1	1	48 287.3587	0.0021
2	2	0	...	2	1	1	7 003.4719	0.0078
2	1	1	...	2	0	2	14 932.2484	0.0010
2	2	1	...	2	1	2	17 712.5383	-0.0074
3	1	3	...	2	0	2	37 792.6326	0.0037
3	0	3	...	2	1	2	37 652.3017	-0.0029

TABLE VI. Rotational constants of four isotopologues of *cis-trans* H₂CO₃.^a

<i>cis-trans</i> H ₂ CO ₃			<i>cis-trans</i> HDCO ₃ ^b		<i>cis-trans</i> DHCO ₃ ^c		<i>cis-trans</i> D ₂ CO ₃	
	Experiment	<i>Ab initio</i> ^d	Experiment	<i>Ab initio</i>	Experiment	<i>Ab initio</i>	Experiment	<i>Ab initio</i>
A	11 778.680 8(35)	11 835	11 764.538 5(97)	11 828	11 608.388 8(84)	11 618	11 290.946 1(64)	11 325
B	11 423.134 5(33)	11 446	10 673.596 2(37)	10 690	10 756.916 6(38)	10 817	10 321.053 6(19)	10 358
C	57 92.074 1(23)	5 818	5 590.123 7(26)	5 615	5 577.018 0(26)	5 602	5 386.688 4(19)	5 410
Δ_J	0.005 73(18)		0.004 86(28)		0.005 39(25)		0.005 54(21)	
Δ_{JK}	−0.001 16(81)		0.002 1(28)		−0.000 5(29)		−0.002 6(20)	
Δ_K	0.008 14(76)		0.006 4(21)		0.007 9(26)		0.009 5(14)	
δ_J	0.0026 18(85)		0.002 618 ^e		0.002 618 ^e		0.002 618 ^e	
δ_K	0.006 28(32)		0.006 28 ^e		0.006 28 ^e		0.006 28 ^e	
σ	10.5 kHz		11.0 kHz		9.08 kHz		6.30 kHz	

^aIn MHz. Values in parentheses denote one standard deviations of the fits.^bThe H atom at the *cis* position and the D atom at the *trans* position.^cThe D atom at the *cis* position and the H atom at the *trans* position.^dComputed at the CCSD(T)/cc-pVQZ level of theory.^eFixed to the values of *cis-trans* H₂CO₃.

definitely shown, in the present study, that carbonic acid can exist in the gas phase, and quantum chemical calculations have predicted isolated carbonic acid is kinetically stable.¹⁴ Thus, carbonic acid may be produced in interstellar and cometary ices and could be released into the gas phase upon mantle evaporation, leading to its detection in the millimeter wavelength region by radio telescopes. Furthermore, it is also expected to exist and may play essential roles in atmospheric chemistry in Earth's atmosphere¹⁵ or atmosphere of other planets,^{34,35} where a certain number of carbonic acid could exist. So far, absence of spectroscopic data had prevented its spectroscopic observations in any areas. We have shown that isolated carbonic acid (*cis-trans* H₂CO₃) exists in the gas phase, and obtained accurate spectroscopic data, which could be used in radio astronomical and atmospheric observations. These observations, if attained, may reveal important roles of carbonic acid in interstellar and atmospheric chemistry.

Given the signal-to-noise ratios of the observed spectra of *cis-trans* H₂CO₃, yet unobserved *cis-cis* H₂CO₃ should be observed by a similar experimental setup, where, however, the small dipole moment of *cis-cis* H₂CO₃ has prevented its observation so far. The CO₂–H₂O complex, carbonic acid, and the complex of carbonic acid with water are key species

for the research of the CO₂ hydration and the carbonic acid dissociation, which are the basic and important reactions in various areas of science. We performed CCSD(T)-F12/aug-cc-pVDZ level of calculations for water complexes of carbonic acid. The most stable geometry of the complex, *cis-cis* H₂CO₃–H₂O, is shown in Fig. 5(a). This complex has two hydrogen bonds with the binding energy calculated to be as large as 11.5 kcal/mol, and the production reaction of carbon dioxide with water is exothermic. This complex could be detected because of its stability and its calculated large dipole moment of $\mu_b=1.05$ D and $\mu_c=1.22$ D. Another complex, the *cis-trans* H₂CO₃–H₂O complex, which is shown in Fig. 5(b), is important as the intermediate of the dissociation process of carbonic acid catalyzed by one water molecule. The transition state structure and the structure of the decomposition product, the H₂O–H₂O–CO₂ complex, are also shown in Figs. 5(c) and 5(d). The H₂O–H₂O–CO₂ complex has already been observed by microwave spectroscopy.³⁶ Calculated energy barrier of this catalyzed dissociation is 27.9 kcal/mol, which is smaller than that of isolated carbonic acid of 49.6 kcal/mol, showing the catalytic effect by one water molecule to the dissociation. Detection of the water complexes of carbonic acid is thus expected to clarify the dissociation dynamics of carbonic acid under the catalytic influence of water.

V. CONCLUSION

Pure rotational transitions of *cis-trans* H₂CO₃ and all the deuterated isotopologues were observed by the FTMW spectrometer and the double resonance technique, which utilizes the FTMW spectrometer and millimeter-wave sources. This is the first detection of isolated carbonic acid, although this molecule has been considered to be a very fundamental and important molecule in various areas of science for a long time, and only quite recently some indirect detection results have been reported. *Ab initio* calculations showed *cis-trans* H₂CO₃ is the second minimum energy structure of carbonic acid, and is an important conformer as the intermediate of the dissociation of *cis-cis* H₂CO₃, which is the global minimum structure of carbonic acid, to CO₂ and H₂O. The r_0

TABLE VII. Geometrical parameters of *cis-trans* carbonic acid. (Values in parentheses denote one standard deviations of the fit.)

	r_0 structure	<i>Ab initio</i> ^a
$r(\text{C}=\text{O})$ (Å)	1.187 88(94)	1.194
$r(\text{C}-\text{O}_{\text{cis}})$ (Å)	1.344 7(13)	1.354
$r(\text{C}-\text{O}_{\text{trans}})$ (Å)	1.356 8(17)	1.336
$\angle(\text{O}=\text{C}-\text{O}_{\text{cis}})$ (deg)	126.78(15)	125.2
$\angle(\text{O}=\text{C}-\text{O}_{\text{trans}})$ (deg)	122.94(12)	124.3
$r(\text{O}_{\text{cis}}-\text{H}/\text{D})$ (Å)	0.968 ^b /0.963 ^c	0.963
$r(\text{O}_{\text{trans}}-\text{H}/\text{D})$ (Å)	0.968 ^b /0.963 ^c	0.963
$\angle(\text{C}-\text{O}_{\text{cis}}-\text{H}/\text{D})$ (deg)	106.1 ^c	106.1
$\angle(\text{C}-\text{O}_{\text{trans}}-\text{H}/\text{D})$ (deg)	108.6 ^c	108.6

^aComputed at the CCSD(T)/cc-pVQZ level of theory.^bFixed to the value of 0.005 Å longer than the *ab initio* value.^cFixed to the *ab initio* value.

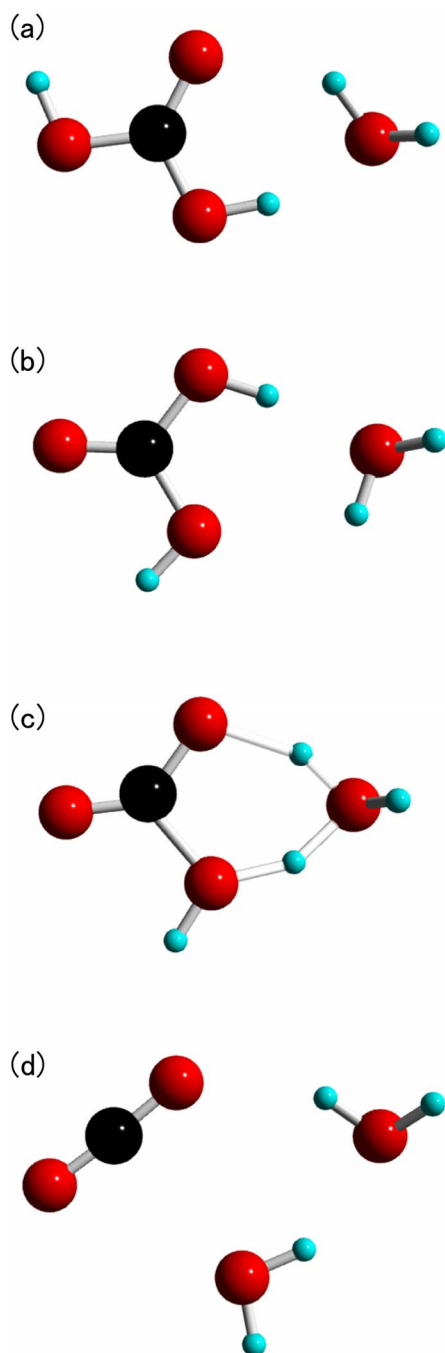


FIG. 5. (Color online) (a) The global minimum structure of the $\text{H}_2\text{CO}_3\text{--H}_2\text{O}$ complex. (b) One of the *cis-trans* $\text{H}_2\text{CO}_3\text{--H}_2\text{O}$ complex, which is important as the starting point structure of the dissociation to $\text{CO}_2 + 2\text{H}_2\text{O}$. (c) Transition state structure of the dissociation from *cis-trans* $\text{H}_2\text{CO}_3\text{--H}_2\text{O}$ complex (b) to $\text{H}_2\text{O--H}_2\text{O--CO}_2$ complex (d). (d) $\text{H}_2\text{O--H}_2\text{O--CO}_2$ complex.

structure of *cis-trans* H_2CO_3 was determined from the rotational constants of four isotopologues. Present results could be utilized to observe carbonic acid in interstellar space by a radio telescope or in Earth's atmosphere.

- ¹D. M. Kern, J. Chem. Educ. **37**, 14 (1960).
- ²H. A. Al-Hosney and V. H. Grassian, J. Am. Chem. Soc. **126**, 8068 (2004).
- ³M. H. Moore and R. K. Khanna, Spectrochim. Acta, Part A **47**, 255 (1991).
- ⁴M. H. Moore, R. L. Hudson, and P. A. Gerakines, Spectrochim. Acta, Part A **57**, 843 (2001).
- ⁵W. Hage, K. R. Liedl, A. Hallbrucker, and E. Mayer, Science **279**, 1332 (1998).
- ⁶K. I. Peterson and W. Klemperer, J. Chem. Phys. **80**, 2439 (1984).
- ⁷J. K. Terlouw, C. B. Lebrilla, and H. Schwarz, Angew. Chem., Int. Ed. Engl. **26**, 354 (1987).
- ⁸K. Winkel, W. Hage, T. Loerting, S. L. Price, and E. Mayer, J. Am. Chem. Soc. **129**, 13863 (2007).
- ⁹B. Jönsson, G. Karlström, H. Wennerström, and B. Roos, Chem. Phys. Lett. **41**, 317 (1976).
- ¹⁰M. T. Nguyen and T.-K. Ha, J. Am. Chem. Soc. **106**, 599 (1984).
- ¹¹R. Janoschek and I. G. Csizmadia, J. Mol. Struct. **300**, 637 (1993).
- ¹²C. A. Wight and A. I. Boldyrev, J. Phys. Chem. **99**, 12125 (1995).
- ¹³M. T. Nguyen, G. Raspoet, L. G. Vanquickenborne, and P. T. van Duijnen, J. Phys. Chem. A **101**, 7379 (1997).
- ¹⁴T. Loerting, C. Tautermann, R. T. Kroemer, I. Kohl, A. Hallbrucker, E. Mayer, and K. R. Liedl, Angew. Chem., Int. Ed. **39**, 891 (2000).
- ¹⁵C. M. Rosado-Reyes and J. S. Francisco, J. Phys. Chem. A **110**, 4419 (2006).
- ¹⁶P. P. Kumar, A. G. Kalinichev, and R. J. Kirkpatrick, J. Chem. Phys. **126**, 204315 (2007).
- ¹⁷T. B. Adler, G. Knizia, and H.-J. Werner, J. Chem. Phys. **127**, 221106 (2007).
- ¹⁸MOLPRO, version 2008.1, a package of *ab initio* programs, H.-J. Werner, P. J. Knowles, R. Lindh, F. R. Manby, M. Schütz *et al.*, see <http://www.molpro.net>.
- ¹⁹M. J. Frisch and G. W. Trucks, H. B. Schlegel *et al.*, GAUSSIAN, a package of *ab initio* program, Gaussian, Inc., Pittsburgh, PA, 2003.
- ²⁰Y. Endo, H. Kohguchi, and Y. Ohshima, Faraday Discuss. **97**, 341 (1994).
- ²¹Y. Hirahara, Y. Ohshima, and Y. Endo, J. Chem. Phys. **101**, 7342 (1994).
- ²²Y. Sumiyoshi, H. Katsunuma, K. Suma, and Y. Endo, J. Chem. Phys. **123**, 054324 (2005).
- ²³J. K. G. Watson, J. Chem. Phys. **46**, 1935 (1967).
- ²⁴J. U. Grabow, W. Stahl, and Z. Naturforsch. A: Phys. Sci. **45**, 1043 (1990).
- ²⁵W. Gordy and R. L. Cook, Microwave Molecular Spectra (Wiley-Interscience, New York, 1970).
- ²⁶G. H. Kwei and R. F. Curl, Jr., J. Chem. Phys. **32**, 1592 (1960).
- ²⁷S.-Y. Liu, J. M. Girart, A. Remijan, and L. E. Snyder, Astrophys. J. **576**, 255 (2002).
- ²⁸H. M. Boechat-Roberty, S. Pilling, and A. C. F. Santos, Astron. Astrophys. **438**, 915 (2005).
- ²⁹W. A. Schutte, L. J. Allamandola, and S. A. Sandford, Science **259**, 1143 (1993).
- ³⁰D. Bockelée-Morvan, D. C. Lis, J. E. Wink, D. Despois, J. Crovisier, R. Bachiller, D. J. Benford, N. Biver, P. Colom, J. K. Davies, E. Gérard, B. Germain, M. Houde, D. Mehringer, R. Moreno, G. Paubert, T. G. Phillips, and H. Rauer, Astron. Astrophys. **353**, 1101 (2000).
- ³¹W. A. Schutte, A. C. A. Boogert, A. G. G. M. Tielens, D. C. B. Whittet, P. A. Gerakines, J. E. Chiar, P. Ehrenfreund, J. M. Greenberg, E. F. van Dishoeck, and Th. de Graauw, Astron. Astrophys. **343**, 966 (1999).
- ³²W. M. Irvine, P. Friberg, N. Kaifu, H. E. Matthews, Y. C. Minh, M. Ohishi, and S. Ishikawa, Astron. Astrophys. **229**, L9 (1990).
- ³³E. F. van Dishoeck, Faraday Discuss. **109**, 31 (1998).
- ³⁴J. S. Lewis and D. H. Grinspoon, Science **249**, 1273 (1990).
- ³⁵G. Strazzulla, J. R. Brucato, G. Cimino, and M. E. Palumbo, Planet. Space Sci. **44**, 1447 (1996).
- ³⁶K. I. Peterson, R. D. Suenram, and F. J. Lovas, J. Chem. Phys. **94**, 106 (1991).